# MATEMATICI 2

## Seminar 9

**Ex.**

Fie $f : \mathbb{R}_2[X] \to \mathbb{R}_2[X]$

$f(P) = P + P'$

$\langle P, Q \rangle = \int_{-1}^{1} P(x)\, Q(x)\, dx$

$f^* = ?$

**Rez.**

Fie $B = \{P_1 = 1, P_2 = X, P_3 = X^2\}$

Cf. ex. ant. $\{Q_1 = 1, Q_2 = X, Q_3 = \frac{X^2 - 1}{3}\}$ este bază ortogonală (cu Gram - Schmidt)

Normalizăm

$$W_1 = \frac{Q_1}{\|Q_1\|} = \frac{1}{\sqrt{\langle Q_1, Q_1 \rangle}} \cdot Q_1 =$$

$$= \frac{1}{\sqrt{\int_{-1}^{1} Q_1(x) Q_1(x)\, dx}} \cdot Q_1 = \frac{1}{\sqrt{2}} \cdot 1 = \frac{1}{\sqrt{2}}$$

$$W_2 = \frac{1}{\|Q_2\|} \cdot Q_2 = \frac{1}{\sqrt{\langle Q_2, Q_2 \rangle}} \cdot Q_2 =$$

$$= \frac{1}{\sqrt{\int_{-1}^{1} Q_2(x) Q_2(x)\, dx}} \cdot Q_2 = \frac{1}{\sqrt{\int_{-1}^{1} x^2\, dx}} \cdot \not{Q}\, x =$$

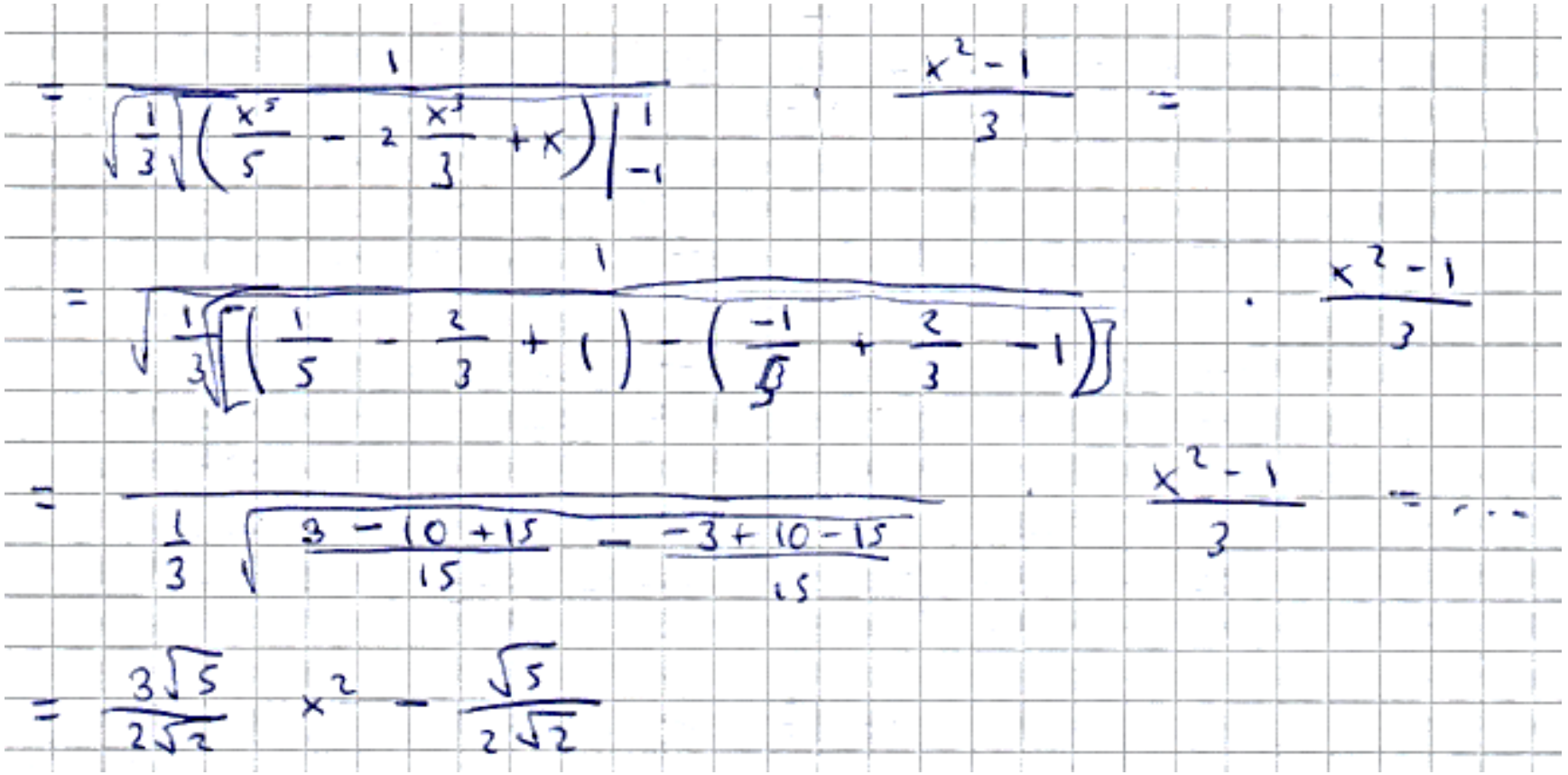
$$= \frac{1}{\sqrt{\frac{x^3}{3}\Big|_{-1}^{1}}} \cdot x = \frac{1}{\sqrt{\frac{1}{3}+\frac{1}{3}}} \cdot x =$$

$$= \sqrt{\frac{3}{2}}\, x$$

$$W_3 = \frac{1}{\|Q_3\|} \cdot Q_3 = \frac{1}{\sqrt{\langle Q_3, Q_3\rangle}} \cdot Q_3 =$$

$$= \frac{1}{\sqrt{\int_{-1}^{1} Q_3(x)\, Q_3(x)\, dx}} \cdot Q_3(x) = \frac{1}{\sqrt{\int_{-1}^{1} \frac{x^2-1}{3} \cdot \frac{x^2-1}{3}\, dx}} \cdot \frac{x^2-1}{3} =$$

$$= \frac{1}{\sqrt{\int_{-1}^{1} \frac{x^4-2x^2+1}{9}\, dx}} \cdot \frac{x^2-1}{3} =$$


$= \frac{1}{\sqrt{\frac{1}{3}\left[\left(\frac{x^5}{5} - 2\frac{x^3}{3} + x\right)\right]_{-1}^{1}}} \cdot \frac{x^2-1}{3} =$
$= \frac{1}{\sqrt{\frac{1}{3}\left[\left(\frac{1}{5} - \frac{2}{3} + 1\right) - \left(\frac{-1}{5} + \frac{2}{3} - 1\right)\right]}} \cdot \frac{x^2-1}{3}$
$= \frac{}{\frac{1}{3}\sqrt{\frac{3-10+15}{15} - \frac{-3+10-15}{15}}} \cdot \frac{x^2-1}{3} = \dots$
$= \frac{3\sqrt{5}}{2\sqrt{2}} x^2 - \frac{\sqrt{5}}{2\sqrt{2}}$

$$B = \left\{ w_1 = \frac{1}{\sqrt{2}},\ w_2 = \sqrt{\frac{3}{2}}\, x,\ w_3 = \frac{3\sqrt{5}}{2\sqrt{2}}\, x^2 - \frac{\sqrt{5}}{2\sqrt{2}} \right\}$$

Fie $f : \mathbb{R}_2[X] \to \mathbb{R}_2[X]$ $\quad f(P) = P + P'$

$\langle P, Q \rangle = \int_{-1}^{1} P(x) \cdot Q(x)\, dx$ $\quad f^* = ?$

$$f(w_1) = \frac{1}{\sqrt{2}} \; ; \; f(w_2) = x\sqrt{\frac{3}{2}} + \sqrt{\frac{3}{2}} = w_2 + \sqrt{3}\, w_1$$

$$f(w_3) = \frac{3\sqrt{5}}{2\sqrt{2}} x^2 - \frac{\sqrt{5}}{2\sqrt{2}} + \frac{3\sqrt{5}}{\sqrt{2}} x = 1 \cdot w_3 +$$

$$f(w_2) = \sqrt{3}\, w_1 + 1 \cdot w_2 + 0 \cdot w_3$$

$$f(w_1) = 1 \cdot w_1 + 0 \cdot w_2 + 0 \cdot w_3$$

$$f(w_3) = 0 \cdot w_1 + \sqrt{5}\, w_2 + 1 \cdot w_3$$

$$M_f^{B_1} = \begin{pmatrix} 1 & \sqrt{3} & 0 \\ 0 & 1 & \sqrt{15} \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow M_{f^*}^{B^*_1} = (M_f^{B_1})^T = \begin{pmatrix} 1 & 0 & 0 \\ \sqrt{3} & 1 & 0 \\ 0 & \sqrt{15} & 1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow f^*(w_1) = 1 \cdot w_1 + \sqrt{3} \cdot w_2 + 0 \cdot w_3$$

$$f^*(w_2) = 0 \cdot w_1 + 1 \cdot w_2 + \sqrt{15} \cdot w_3$$

$$f^*(w_3) = 0 \cdot w_1 + 0 \cdot w_2 + 1 \cdot w_3$$

Căutăm $\alpha, \beta, \gamma$ a.î. $ax^2+bx+c = \alpha u_1 + \beta u_2 + \gamma u_3$

$$ax^2+bx+c = \frac{\alpha}{\sqrt{2}} + \sqrt{\frac{3}{2}}\beta x + \frac{3\sqrt{5}}{2\sqrt{2}}\gamma^2 x^2 - \frac{\sqrt{5}}{2\sqrt{2}}\gamma$$

$$a = \frac{3\sqrt{5}}{2\sqrt{2}}\gamma \Rightarrow \gamma = \frac{3\sqrt{5}}{2\sqrt{2}a}\cdot\frac{2\sqrt{2}}{3\sqrt{5}}a$$

$$b = \sqrt{\frac{3}{2}}\beta \Rightarrow \beta = \sqrt{\frac{3}{2}}\cdot\frac{1}{b} \qquad \beta = \sqrt{\frac{2}{3}}b$$

$$\frac{2\sqrt{2}}{3\sqrt{5}}a$$

$$c = \frac{1}{\sqrt{2}} - \frac{\sqrt{5}}{2\sqrt{2}}\gamma = \frac{1}{\sqrt{2}} - \frac{5}{2\sqrt{2}}\cdot\frac{3\sqrt{5}}{2\sqrt{2}a} \Rightarrow$$

$$\Rightarrow a = \left(c + \frac{15\sqrt{5}}{8a}\right)\sqrt{2}\,2\sqrt{2}\,\frac{\sqrt{5}}{2\sqrt{2}}\cdot\frac{2\sqrt{2}}{3\sqrt{5}}a\Big)\sqrt{2}$$

$$\frac{\sqrt{5}}{3}a$$

$$ax^2+bx+c=\left(c+\frac{15\sqrt{5}}{8a}\right)\sqrt{2}\,w_1+\sqrt{\frac{3}{2}}\cdot\frac{1}{b}\,w_2+$$

$$+\frac{3\sqrt{5}}{2\sqrt{2}a}\,w_3$$

$$\sqrt{\frac{2}{3}b}$$

$$\frac{2\sqrt{2}}{3\sqrt{5}}$$

$$ax^2+bx+c=\left(c+\frac{\sqrt{5}}{3}a\right)\sqrt{2}\,w_1+\sqrt{\frac{2}{3}b}\,w_2+$$

$$+\frac{2\sqrt{2}}{3\sqrt{5}}\,w_3$$

Înlocuim $u_1, u_2, u_3$.

$$f^*(ax^2+bx+c) = f^*\left(\left(c+\frac{\sqrt{3}}{2}a\right)\sqrt{2}\,u_1\right) +$$

$$+ f^*\left(\sqrt{\frac{2}{3}}b\,u_2\right) + f^*\left(\frac{2\sqrt{2}}{3\sqrt{5}}u_3\right)$$

$$f^*(ax^2+bx+c) = \left(c+\frac{\sqrt{3}}{2}a\right)\sqrt{2}\, f^*(w_1) +$$

$$+\sqrt{\frac{2}{3}}\, b\, f^*(w_2) + \frac{2\sqrt{2}}{3\sqrt{5}} f^*(w_3)$$

Înlocuim $f^*(w_i)$, $i=\overline{1,3}$ și

după înlocuim $w_i$, $i=\overline{1,3}$

**Ex.** Să se det. câte matrici ortogonale cu elemente din $\mathbb{Z}$ există?

Fie $A \in \mathcal{M}_3(\mathbb{Z})$, $A$ = ortogonală $\Leftrightarrow A^T = A^{-1}$

## Rez.

Cons. $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ $A^T = \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix}$

$A \cdot A^T = A \cdot A^{-1} = I_3$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} = \begin{pmatrix} a^2+b^2+c^2 & ad+be+cf & ag+bh+ci \\ ad+be+cf & d^2+e^2+f^2 & dg+eh+fi \\ ag+bh+ci & dg+eh+fi & g^2+h^2+i^2 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{cases} a^2 + b^2 + c^2 = 1 \\ d^2 + e^2 + f^2 = 1 \\ g^2 + h^2 + i^2 = 1 \\ ad + be + cf = 0 \\ ag + bh + ci = 0 \\ dg + eh + fi = 0 \end{cases} \quad (*)$$

$a, b, c, d, e, f, g, h, i \in \mathbb{Z} \Rightarrow$

$\Rightarrow (a,b,c) \in \{(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)\}$

$(d,e,f) \in$ -"-

$(g,h,i) \in$ -"-

$\Rightarrow$ (*) Pe fiecare linie / col. se află un SINGUR elem. $\pm 1$!

$f_3$ $\quad 3! \cdot 2^3 = 2 \cdot 3 \cdot 8 = 6 \cdot 8 = 48$

**Ex.** Fie $f : \mathbb{R}^3 \to \mathbb{R}^3$.

$f(x, y, z) = (x + 2y + z,\ 2x + y + 3z,\ x + 3y + z)$

Să se arate că $f = f^*$

Caus. $\langle (x_1, y_1, z_1), (x_2, y_2, z_2) \rangle =$

$= x_1 x_2 + y_1 y_2 + z_1 z_2$

**Rez.** Vom arăta că $M_{f^*}^{B} = M_f^{B}$

$B = \{e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)\}$ bază

$f(e_1) = (1,2,1) = 1 \cdot e_1 + 2 \cdot e_2 + 1 \cdot e_3$

$f(e_2) = (2,1,3) = 2 \cdot e_1 + 1 \cdot e_2 + 3 \cdot e_3$

$f(e_3) = (1,3,1) = 1 \cdot e_1 + 3 \cdot e_2 + 1 \cdot e_3$

$$M_f^{\mathcal{B}} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 3 & 1 \end{pmatrix}$$

$$M_{f^*}^{\mathcal{B}} = (M_f^{\mathcal{B}})^T = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 3 & 1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow M_{f^*}^{\mathcal{B}} = M_f^{\mathcal{B}} \Rightarrow f = f^*$$

**Ex.** $f : \mathcal{M}_2(\mathbb{R}) \to \mathcal{M}_2(\mathbb{R})$

$$f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & b \\ c & a \end{pmatrix}$$

$f = f^*$

$$\langle A, B \rangle = \sum_{i=1}^{2} \sum_{j=1}^{2} a_{ij} b_{ij}$$

Cons. $B = \left\{ E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \right.$

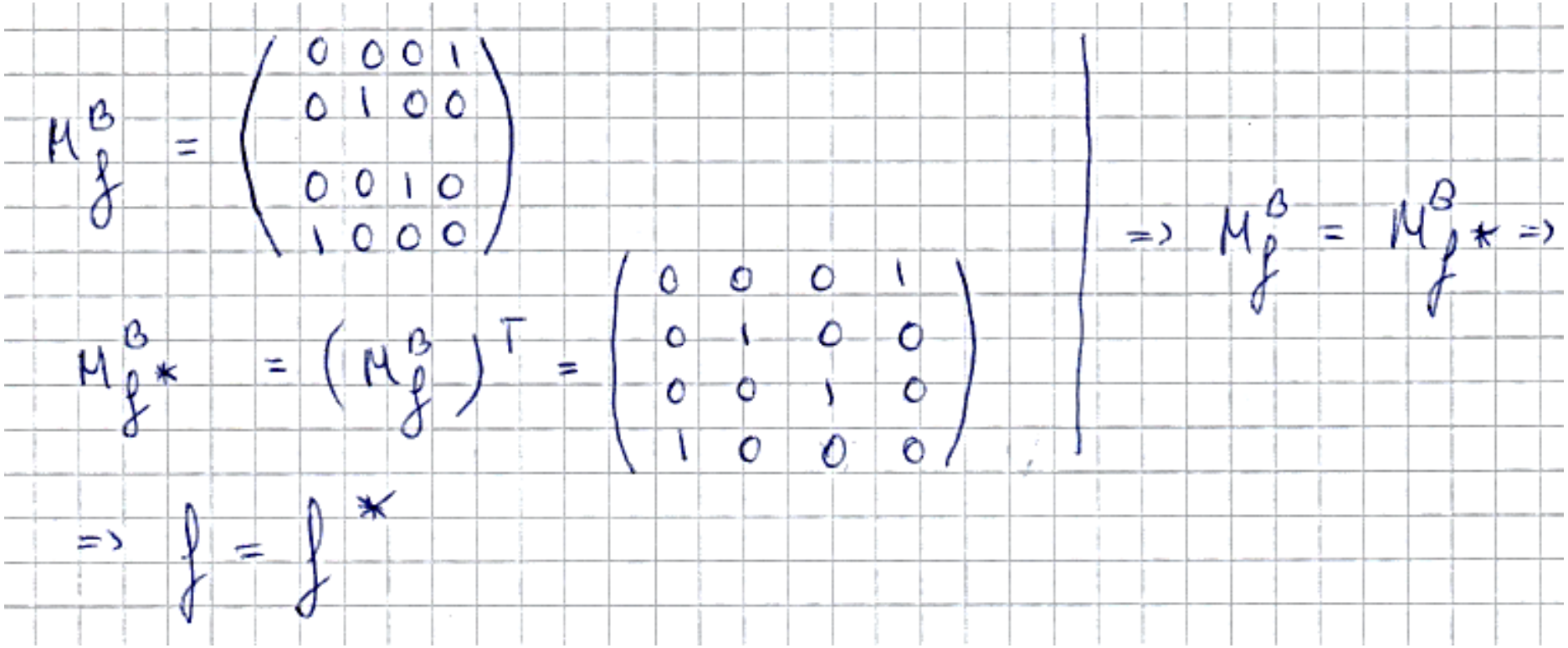
$\left. E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$

$f(E_{11}) = f\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = E_{22} = 0 \cdot E_{11} + 0 \cdot E_{12} + 0 \cdot E_{21} + 1 \cdot E_{22}$

$f(E_{12}) = f\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = E_{12} = 0 \cdot E_{11} + 1 \cdot E_{12} + 0 \cdot E_{21} + 0 \cdot E_{22}$

$$f(E_{21}) = f\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = E_{21} = 0 \cdot E_{11} + 0 \cdot E_{12} + 1 \cdot E_{21} + 0 \cdot E_{22}$$

$$f(E_{22}) = f\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = E_{11} = 1 \cdot E_{11} + 0 \cdot E_{12} + 0 \cdot E_{21} + 0 \cdot E_{22}$$


$M_f^B = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$
$M_{f^*}^B = (M_f^B)^T = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$
$\Rightarrow M_f^B = M_{f^*}^B \Rightarrow$
$\Rightarrow f = f^*$